Time series: SSA

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The homework is provided on the last slide

Singular Spectrum Analysis (SSA)

SSA is a model-free methodology for the analysis of time series.

SSA is applied to a wide range of problems including

- Discovery of structure.
- Signal/trend extraction.
- Smoothing and noise reduction.
- Extraction of seasonal or harmonic components.
- Forecasting.
- Change-point detection.

SSA software `http://www.gistatgroup.com/cat/`
R-package `http://cran.r-project.org/web/packages/Rssa/`
SSA with R book `http://ssa-with-r-book.github.io`

SSA algorithm: initial stage

The initial step of the SSA algorithm is a transformation of a time series into a matrix.

f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8	f_9

Suppose that we are given a time series $f_1, \ldots, f_9$. We convert the given vector (of time series) into a matrix $\mathbf{X}$ using a sliding window and writing elements from this window as columns of the matrix $\mathbf{X}$.

Consider a window length of 3.

Transformation of data: column 1

f_1	f_2	f_3

f_4 f_5 f_6 f_7 f_8 f_9

f_1
f_2
f_3

Transformation of data: column 2

f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8	f_9

f_1	f_2
f_2	f_3
f_3	f_4

Transformation of data: column 3

$$f_1 \quad f_2 \quad \boxed{f_3 \mid f_4 \mid f_5} \quad f_6 \quad f_7 \quad f_8 \quad f_9$$

f_1	f_2	f_3
f_2	f_3	f_4
f_3	f_4	f_5

Transformation of data: column 4

$$f_1 \quad f_2 \quad f_3 \quad \boxed{f_4} \quad \boxed{f_5} \quad \boxed{f_6} \quad f_7 \quad f_8 \quad f_9$$

f_1	f_2	f_3	f_4
f_2	f_3	f_4	f_5
f_3	f_4	f_5	f_6

Transformation of data: column 5

$$f_1 \quad f_2 \quad f_3 \quad f_4 \quad \boxed{f_5} \quad \boxed{f_6} \quad \boxed{f_7} \quad f_8 \quad f_9$$

f_1	f_2	f_3	f_4	f_5
f_2	f_3	f_4	f_5	f_6
f_3	f_4	f_5	f_6	f_7

Result of transformation: trajectory matrix

f_1	f_2	f_3	f_4	f_5	f_6	f_7	$\cdots$	f_n

$= F$

f_1	f_2	f_3	$\cdots$	f_{n-L+1}
f_2	f_3	f_4	$\cdots$	f_{n-L+2}
f_3	f_4	f_5	$\cdots$	f_{n-L+3}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
f_L	f_{L+1}	f_{L+2}	$\cdots$	f_n

$= \mathbf{X}$

$\mathbf{X}$ is called a trajectory matrix, $2 \leq L < n - L + 1$

Decomposition of trajectory matrix

Singular Value Decomposition (SVD)

$$\mathbf{X} = \sqrt{\lambda_1} U_1 V_1^T + \sqrt{\lambda_2} U_2 V_2^T + \sqrt{\lambda_3} U_3 V_3^T + \ldots$$

- (λ_i, U_i, V_i) is called an eigentriple
- λ_i is an eigenvalue of $\mathbf{X}\mathbf{X}^T$, $i = 1, \ldots, L$
- U_i is called an eigenfunction
- V_i is called a factor vector
- $\dfrac{\lambda_i}{\sum_{j=1}^{L} \lambda_j}$ is the proportion of variance explained by the ith component

Decomposition of series

$$\mathbf{X} = \sqrt{\lambda_1} U_1 V_1^T + \sqrt{\lambda_2} U_2 V_2^T + \sqrt{\lambda_3} U_3 V_3^T + \dots$$

$$\Uparrow \quad \circlearrowleft \quad \Downarrow \qquad \Downarrow \qquad \Downarrow$$

$$F = F_1 + F_2 + F_3 + \dots$$

- F_i is a trend component
- F_j is a harmonic component
- F_l is a noise component

Elements of SSA theory

The equivalence of three definitions

- Series of finite rank
- Series of finite order
- Series of finite difference dimension

1. Series of finite rank

Definition

A time series F has a rank d if the dimension of the linear space spanned by the columns of the trajectory matrix is d for any $L > d$ and $N - L > d$, i.e. $\mathrm{rank}(F) := \mathrm{rank}(\mathbf{X})$.

$$\mathbf{X} = \begin{array}{|ccccc|} \hline f_1 & f_2 & f_3 & \cdots & f_{n-L+1} \\ f_2 & f_3 & f_4 & \cdots & f_{n-L+2} \\ f_3 & f_4 & f_5 & \cdots & f_{n-L+3} \\ \vdots & \vdots & \vdots & & \vdots \\ f_L & f_{L+1} & f_{L+2} & \cdots & f_n \\ \hline \end{array}$$

2. Series of finite order

Definition

A time series F has a order d if there exist two systems of functions

$$\begin{aligned} \phi_1, \ldots, \phi_d &: \{1, \ldots, L\} \to \mathbb{R}, \\ \psi_1, \ldots, \psi_d &: \{1, \ldots, N-L+1\} \to \mathbb{R} \end{aligned}$$

such that

$$f_{i+j} = \sum_{k=1}^{d} \phi_k(i)\psi_k(j)$$

for any $i = 1, \ldots, L,\ j = 1, \ldots, N-L+1$.

Example. Let $f_i = a^i$. Then $\mathrm{ord}(F) = 1$ since

$$f_{i+j} = a^{i+j} = \phi_1(i)\psi_1(j),$$

$\phi_1(i) = a^i$ and $\psi_1(j) = a^j$.

3. Series of finite difference dimension

Definition

A time series F has a finite difference dimension d if there are numbers $\alpha_1, \ldots, \alpha_d$ such that $\alpha_d \neq 0$ and the series F satisfies the linear recurrence formula with coefficients $\alpha_1, \ldots, \alpha_d$, that is

$$f_{i+d} = \sum_{k=1}^{d} \alpha_k f_{i+d-k}$$

for any $i = 1, \ldots, n - d$.

Example. Let $f_i = a^i$. Then $\dim(F) = 1$ since $f_{i+1} = \alpha f_i$ for $\alpha = a$.

Results from theory of diff. equations

- Ordinary differential equation of order d with respect to $y = y(t)$

$$y^{(n)} + c_{n-1}y^{(n-1)} + \ldots + c_1 y' + c_0 y = 0.$$

- The characteristic polynomial has n roots, $\gamma_k \in \mathbb{C}$

$$z^n + c_{n-1}z^{n-1} + \ldots + c_1 z + c_0 = \prod_{k=1}^{h} (z - \gamma_k)^{m_k}$$

and m_k is the corresponding multiplicities.

- Then the functions

$$y_{k,j}(t) = t^j e^{-\gamma_k t}, \ j = 0, \ldots, m_k - 1, \ \kappa = 1, \ldots, h$$

are n linearly independent solutions of ODE.

Practical value of SSA

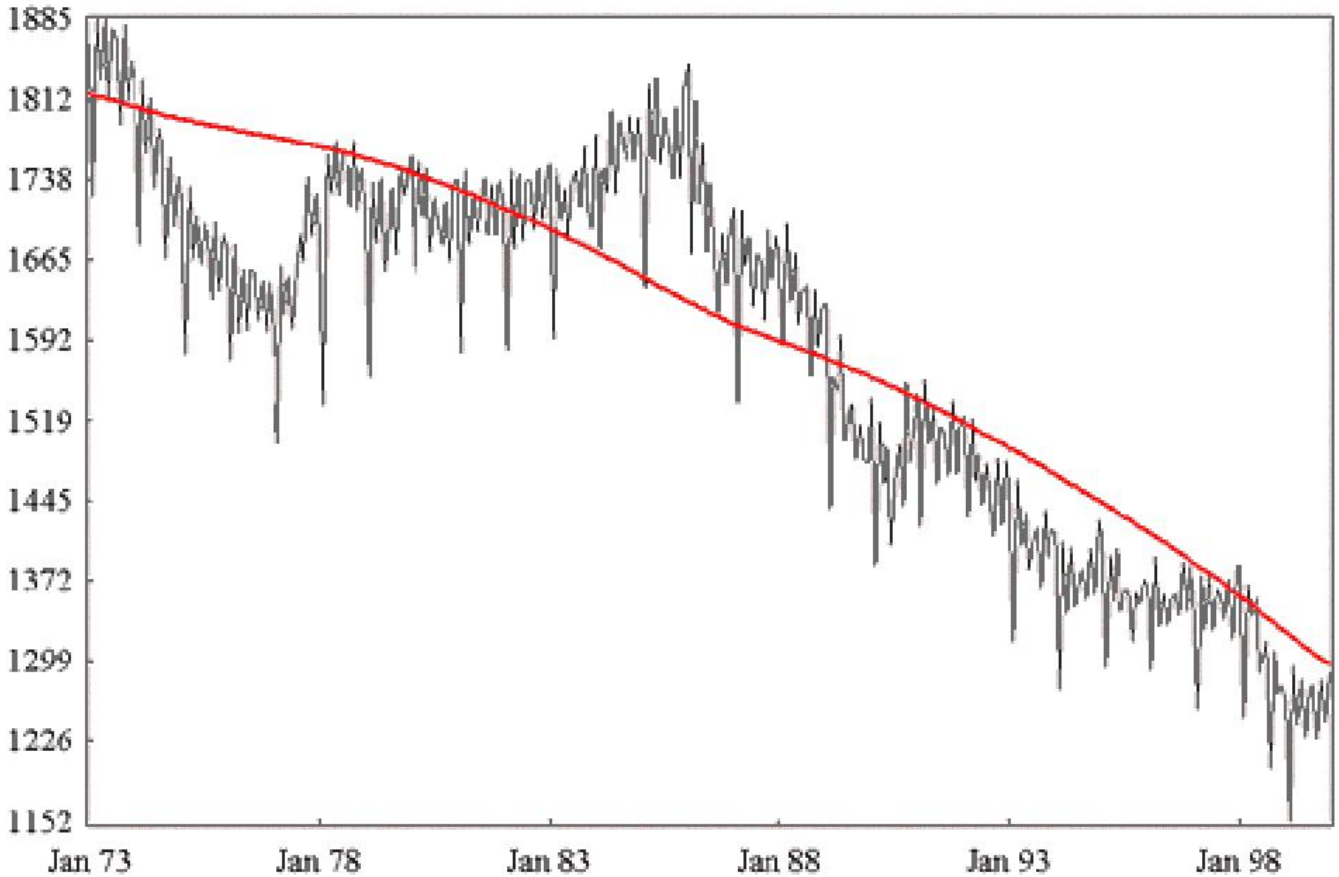
- The series given by

$$f_t = \sum_k P_k(t) e^{-\mu_k t} \sin(\alpha_k t + \beta_k), \ t = 1, \ldots, n$$

  is a series of finite rank (and of finite order...).
- SVD decomposition of $\mathbf{X}$ for a time series of finite rank d has only d nonzero eigenvalues.

- A real time series is a sum of a series of finite rank and a noise series.
- SSA provides good results in more general cases.

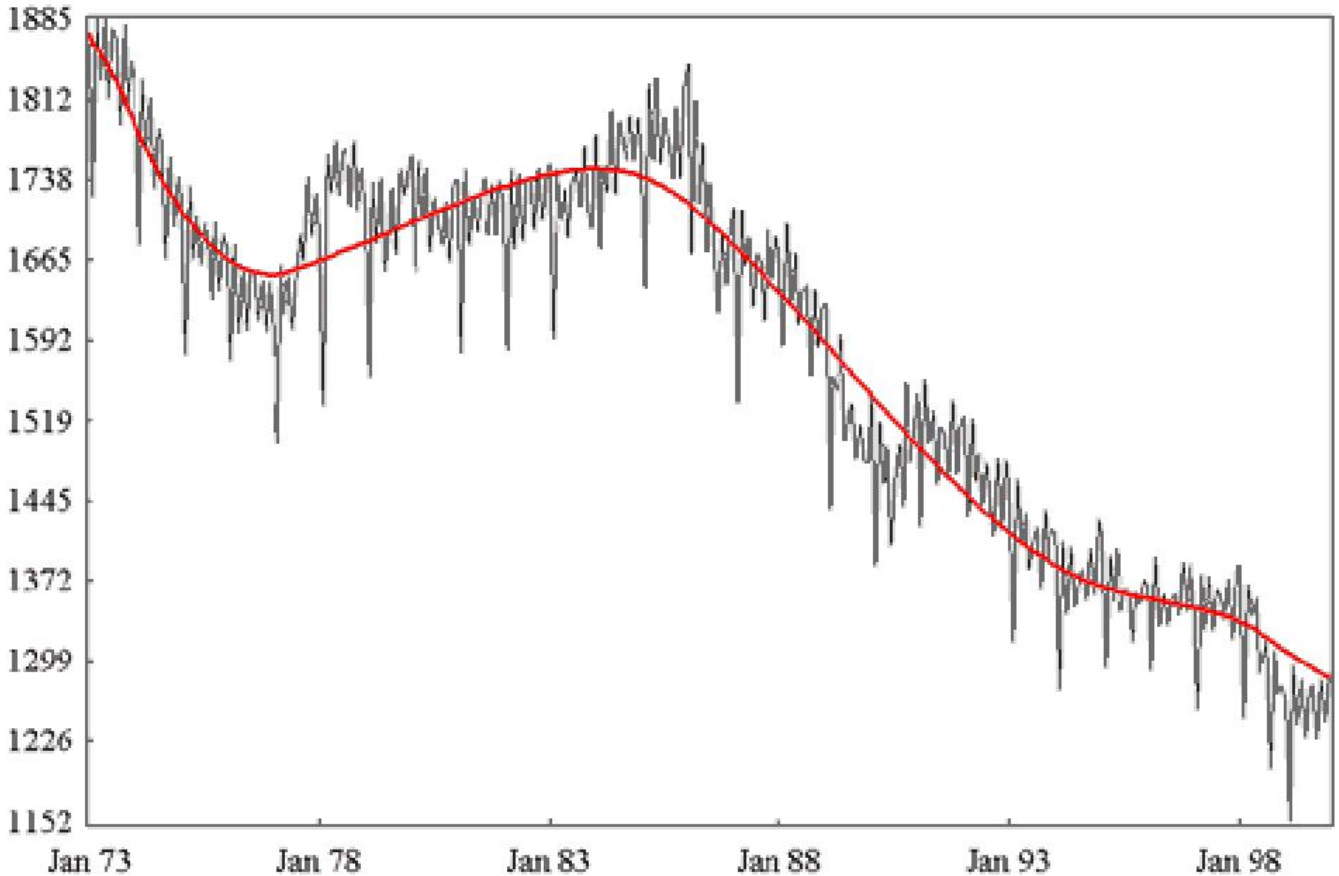
Rough trend extraction



Crude oil production, monthly data from Jan 1973 to Sep 1999

Reconstruction from the leading eigentriple, $L = 120$, $n = 324$

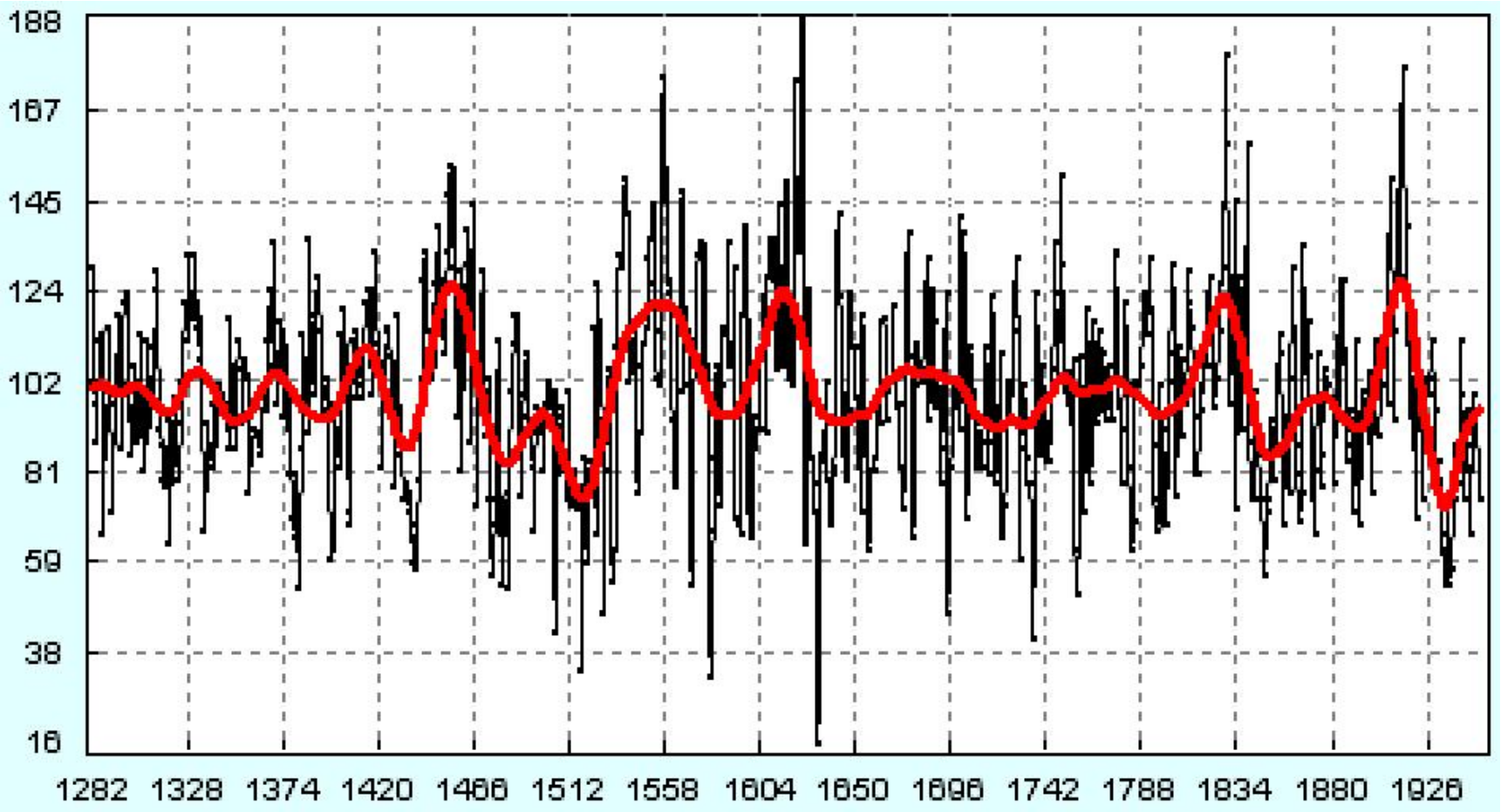
Fine trend extraction



Crude oil production, monthly data from Jan 1973 to Sep 1999

Reconstruction from 3 leading eigentriples, $L = 120$, $n = 324$

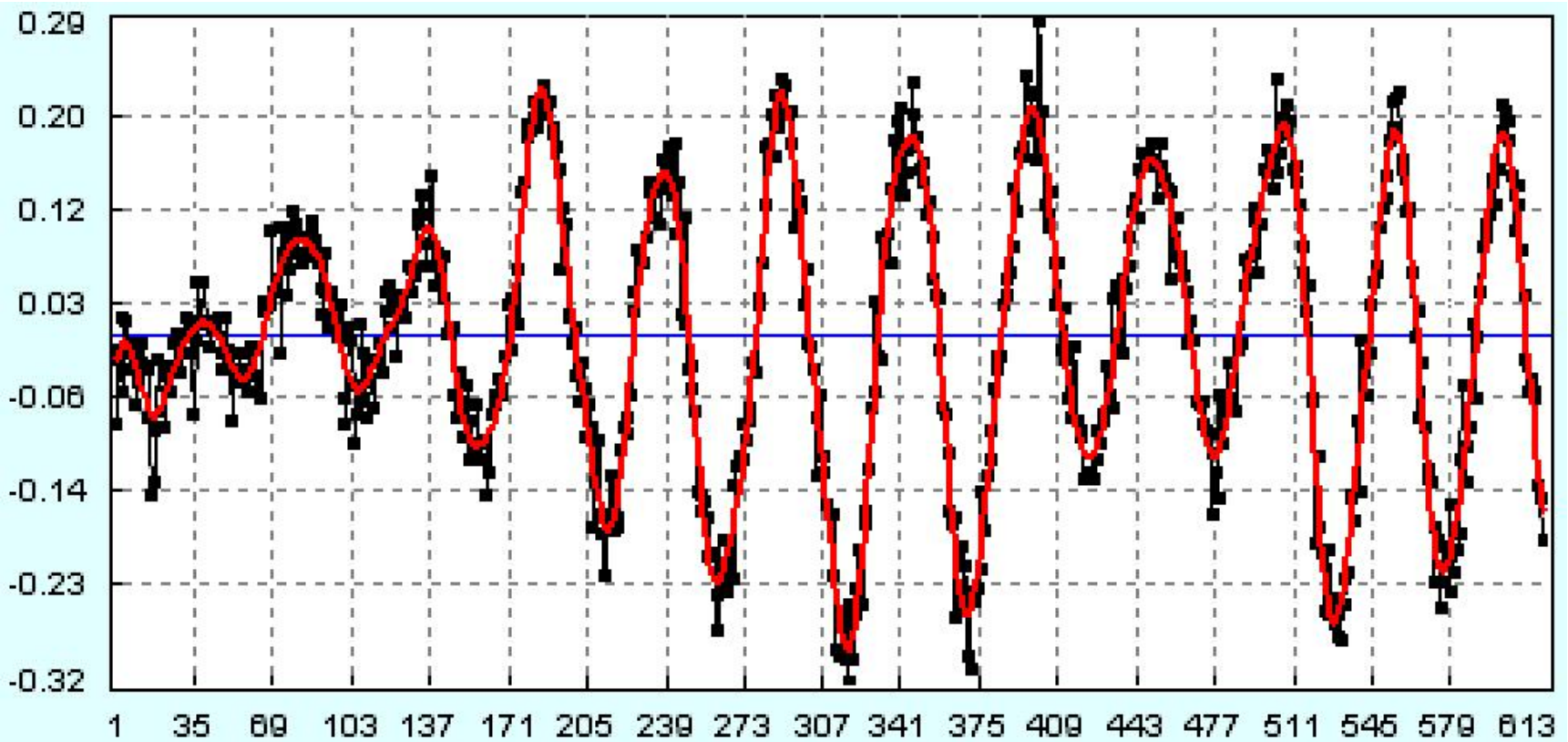
Smoothing



Tree ring indices, Douglas fir, annual, from 1282 to 1950

Reconstruction from 7 leading eigentriples, $L = 120$, $n = 669$

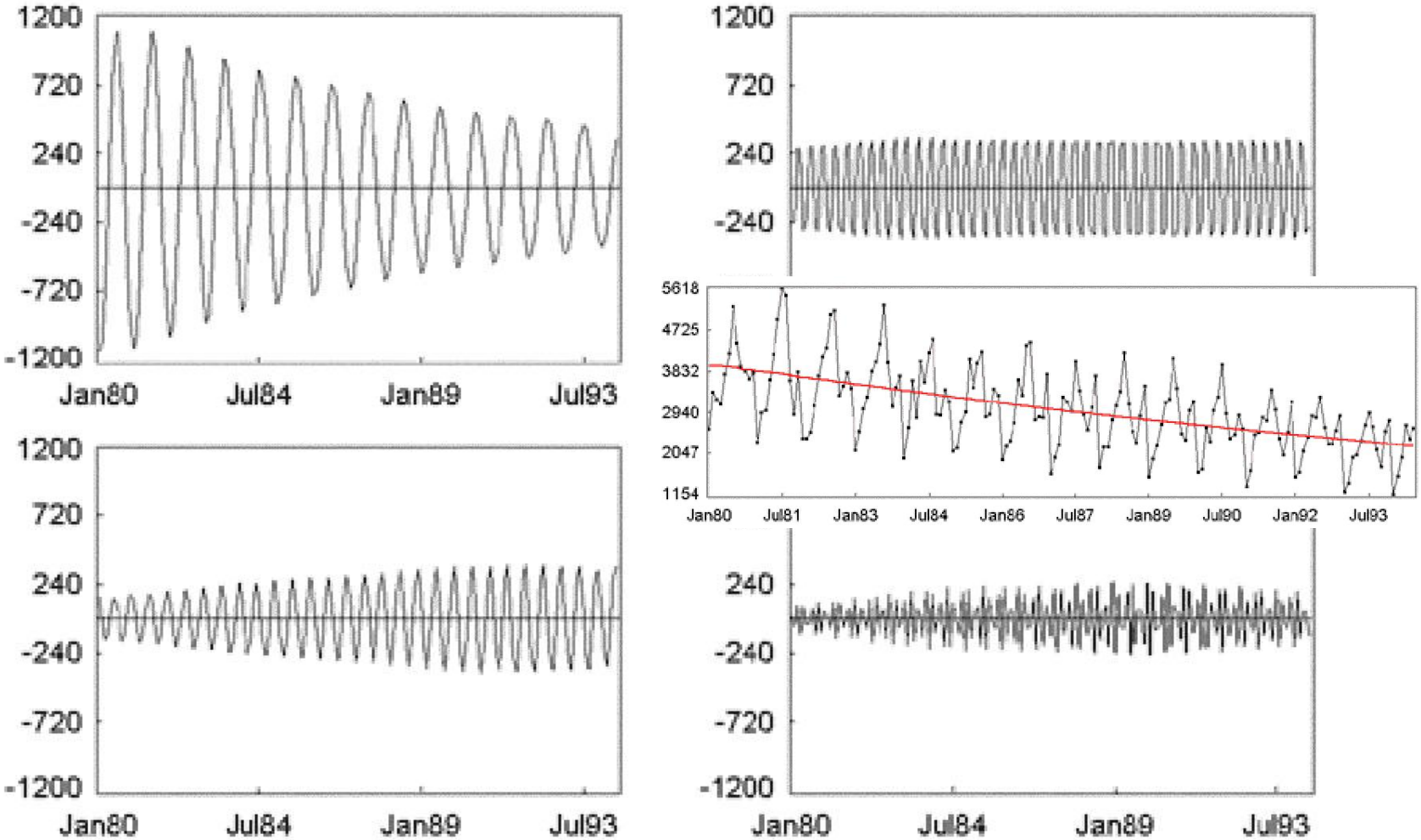
Noise reduction



Intensity of the white dwarf star PG1159-035 during March 1989

Reconstruction from 11 leading eigentriples, $L = 100$, $n = 618$

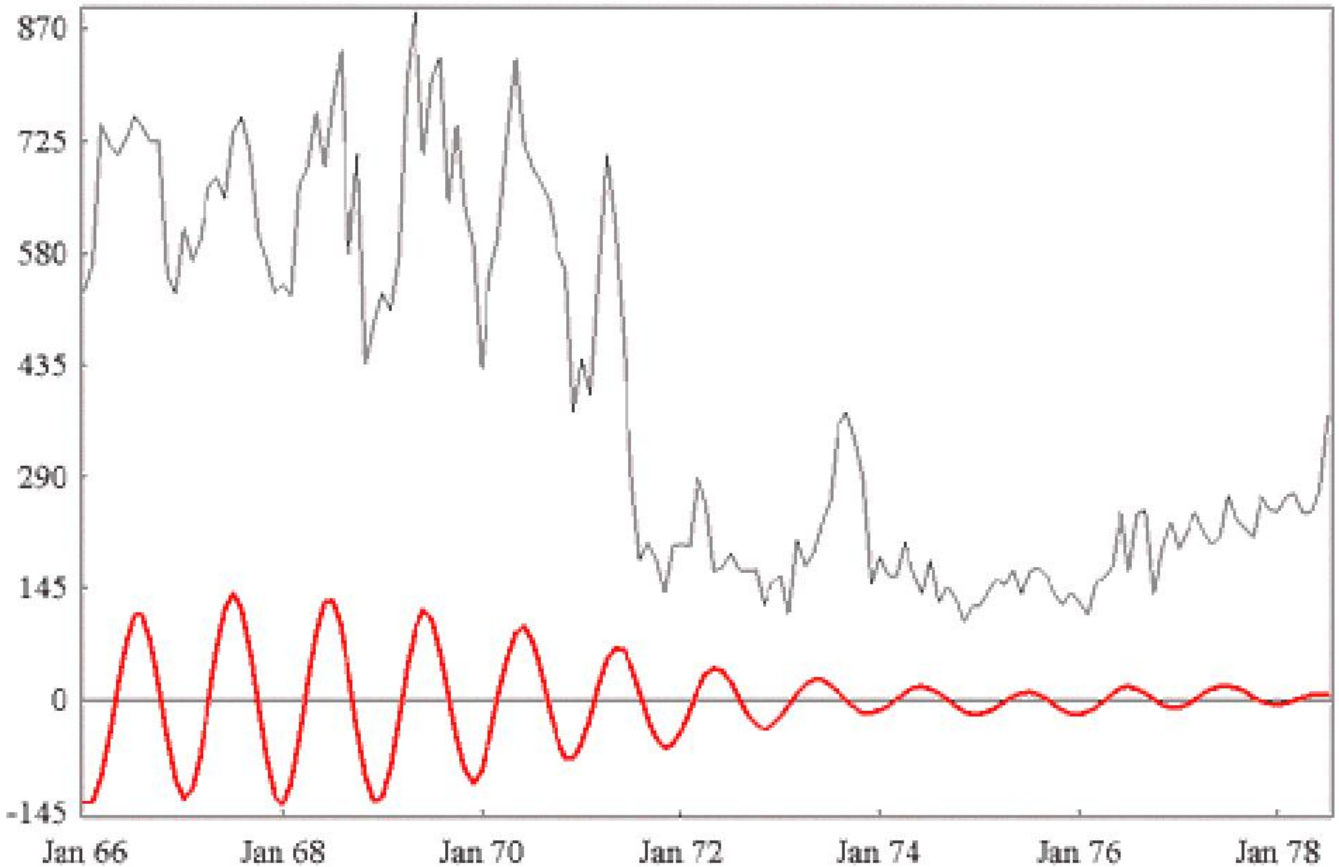
Extraction of seasonality



Sales of fortified wines in Australia, monthly, Jan 1980-Jun 1994

Reconstruction from eigentriples 2-3, 4-5, 6-7, 8-9, $L = 84$, $n = 174$

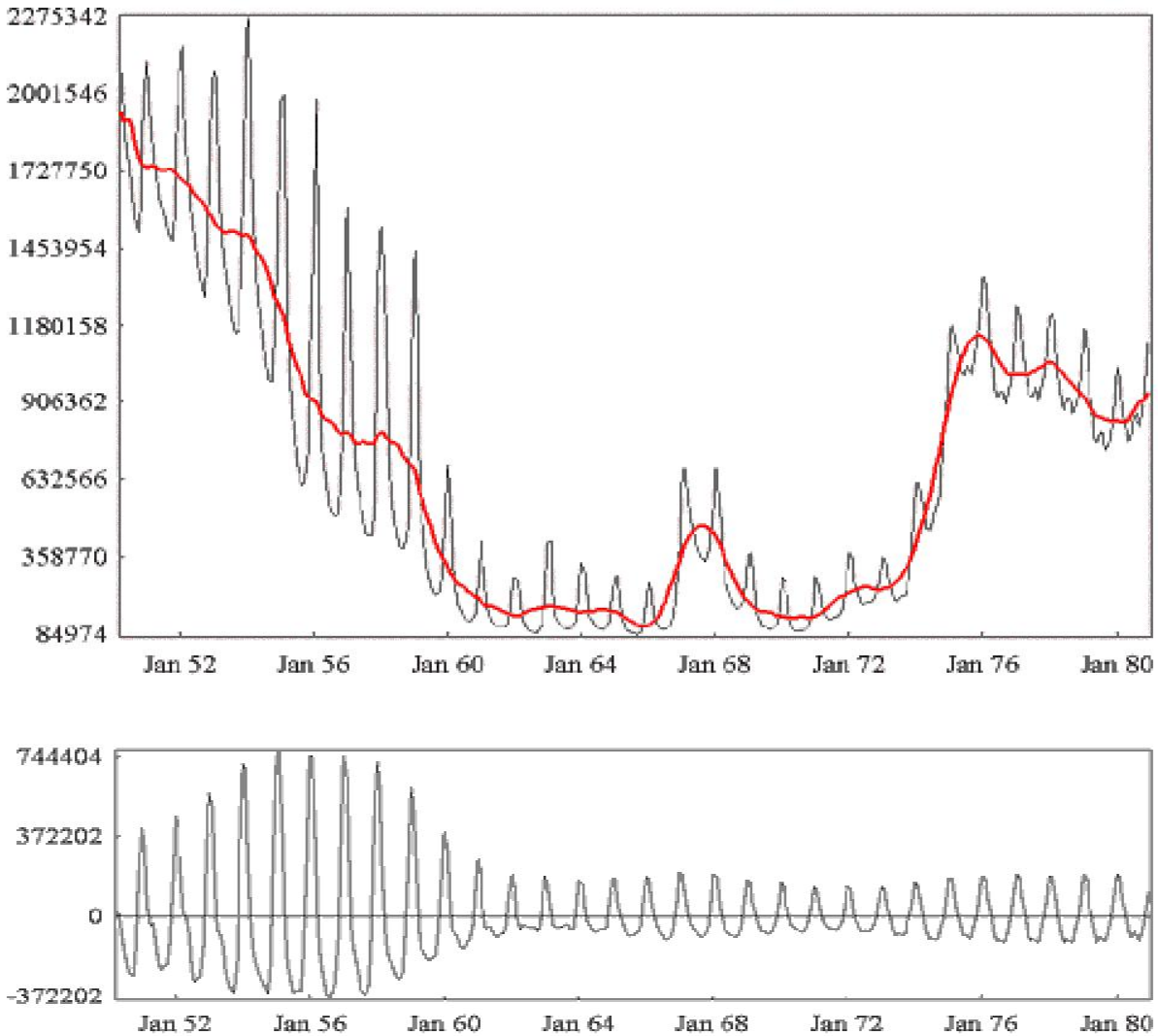
Extraction of periodicities with varying amplitudes



Monthly public drunkenness intakes, from Jan 1966 to Jul 1978

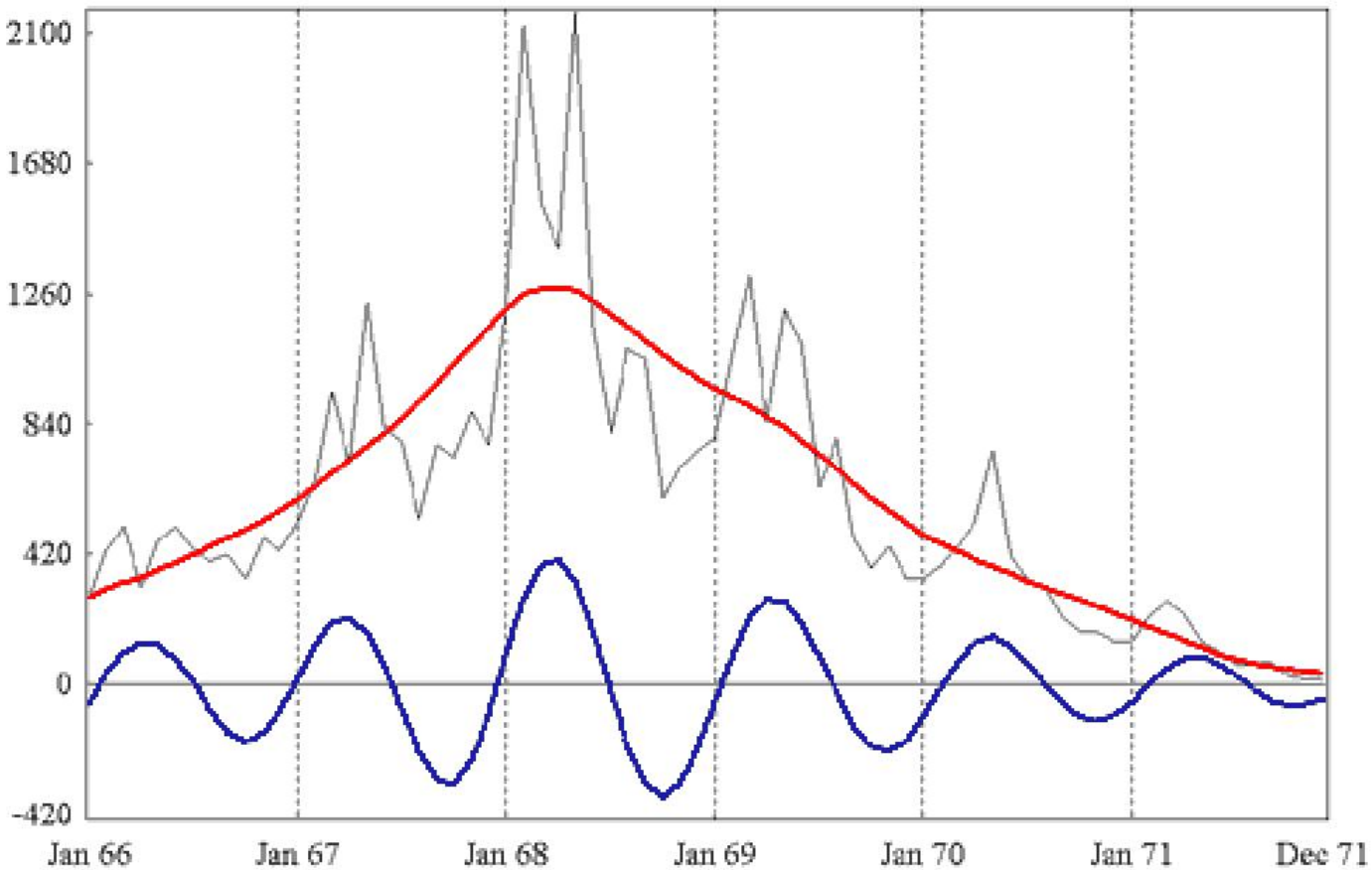
Reconstruction from the 4th-5th eigentriples, $L = 60$, $n = 151$

Complex trends and periodicities



Unemployment, West Germany, monthly, from Apr 1950 to Dec 1980

Finding structure in short time series



Deaths in the Indochina war, monthly, from 1966 to 1971

Reconstruction from eigentriples 1-2, 3-4, $L = 18$, $n = 72$

Methods of forecasting

There are two methods of SSA forecasting

- Recurrent forecasting
- Vector forecasting

Recurrent forecasting

Let signal satisfy LRF $s_j = \alpha_1 s_{j-1} + \ldots + \alpha_q s_{j-q}$.

- Construct LRF
  $\tilde{s}_j = -(a_1 \tilde{s}_{j-1} + \ldots + a_{L-1} \tilde{s}_{j-L+1})/a_L$

  $$(a_L, \ldots, a_1)^T = \mathbf{U}_{r+1,L} \mathbf{U}_{r+1,L}^T e_L$$

  $\mathbf{U}_{r+1,L} = (U_{r+1} \vdots \ldots \vdots U_L)$, $e_L = (0, \ldots, 0, 1)^T \in \mathbb{R}^L$

  $U_{r+1}, \ldots, U_L$ are orthogonal to the signal subspace.
- Apply this LRF to $\hat{S}$ recursively.

Vector forecasting

$$\mathbf{X}_{(s)} = \sum_{i=1}^{r} \sqrt{\lambda_i} U_i V_i^T = \begin{array}{|ccccc|} \hline x_{11} & x_{12} & x_{13} & \cdots & x_{1K} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2K} \\ x_{31} & x_{32} & x_{33} & \cdots & x_{3K} \\ \vdots & \vdots & \vdots & & \vdots \\ x_{L1} & x_{L2} & x_{L3} & \cdots & x_{LK} \\ \hline \end{array}$$

- $K = N - L + 1$.
- rank$\mathbf{X}_{(s)} = r$.
- Columns of $\mathbf{X}_{(s)}$ generate span the signal subspace.
- $U_1, \ldots, U_r$ is a basis of this signal subspace.

Vector forecasting

x_{11}	x_{12}	x_{13}	$\cdots$	x_{1K}	z_1
x_{21}	x_{22}	x_{23}	$\cdots$	x_{2K}	z_2
x_{31}	x_{32}	x_{33}	$\cdots$	x_{3K}	z_3
$\vdots$	$\vdots$	$\vdots$		$\vdots$	
x_{L1}	x_{L2}	x_{L3}	$\cdots$	x_{LK}	z_L

Add a new column such that it belongs to the signal subspace

- $(z_1, z_2 \ldots, z_{L-1})^T$ is the orthogonal projection of $(x_{2K}, x_{3K}, \ldots, x_{LK})^T$
- $z_L = -(a_1 z_{L-1} + \ldots + a_{L-1} z_1)/a_L$

Vector forecasting

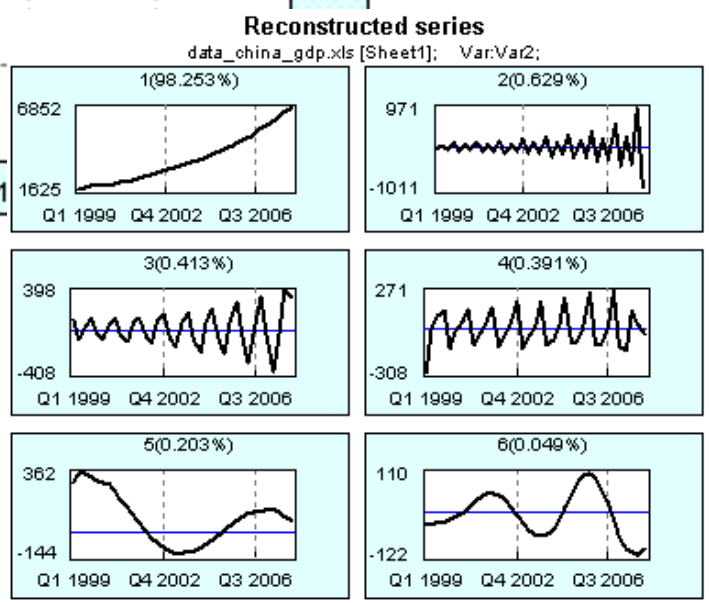
x_{11}	x_{12}	x_{13}	$\cdots$	x_{1K}	z_1			
x_{21}	x_{22}	x_{23}	$\cdots$	x_{2K}	z_2			
x_{31}	x_{33}	x_{33}	$\cdots$	x_{3K}	z_3			
$\vdots$	$\vdots$	$\vdots$		$\vdots$				
x_{L1}	x_{L2}	x_{L3}	$\cdots$	x_{LK}	z_L			

$\underbrace{\qquad\qquad}_{L}$

- To make h-step ahead forecast, we need to add $h + L$ columns
- Perform hankelization and then remove last L elements to obtain forecast

Example of forecast: China GDP





ET 1 is a trend

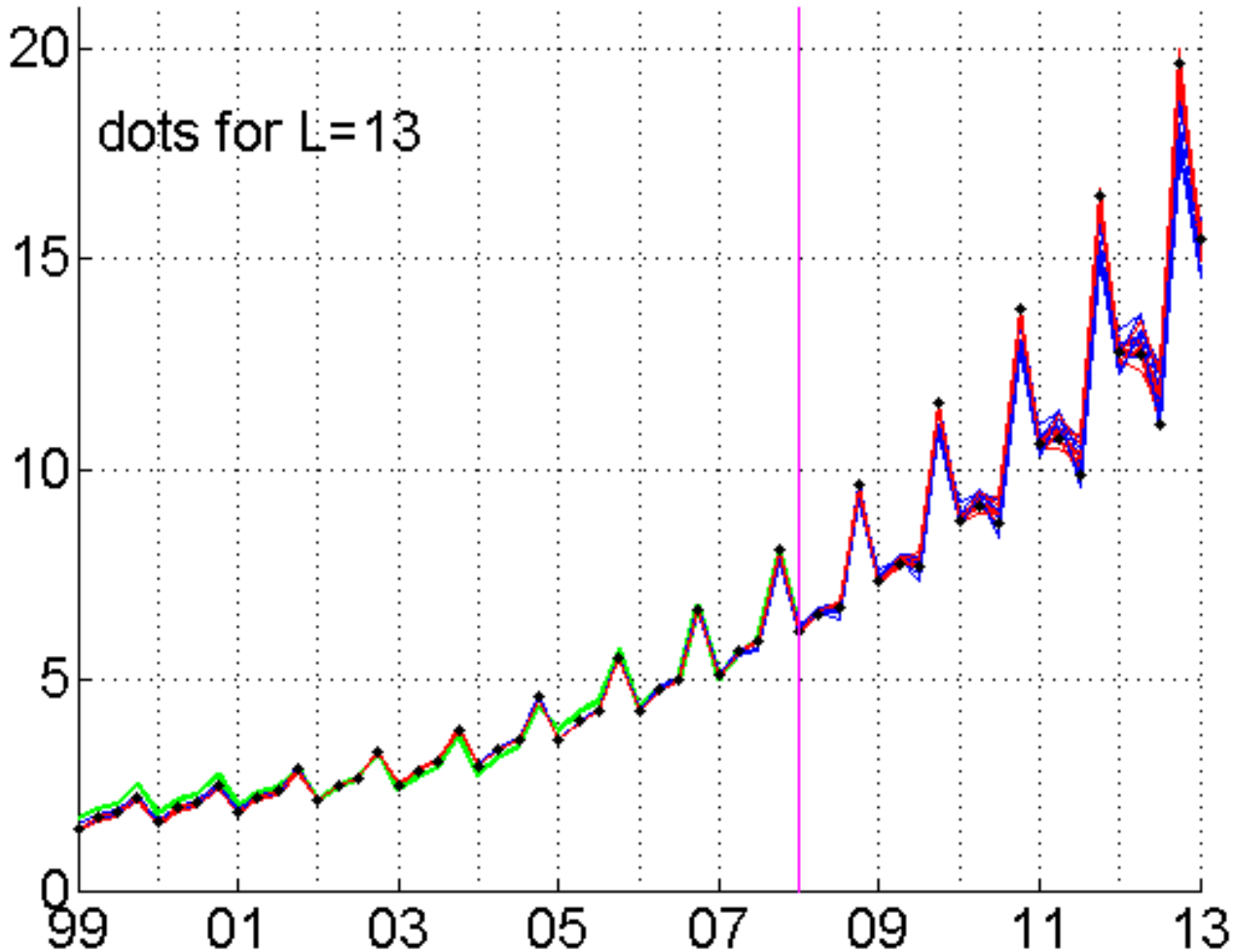
ET 2 is $\approx$ a harmonic with $T = 2$

ET 3&4 are harmonics with $T = 4$

ET 5 is a minor correction to trend

Forecasts of China GDP, $r = 4$

GDP



Recurrent (red) and vector (blue) forecasts, $L = 12, ..., 19$

R-code for time series of CO_2

®

```
install.packages("Rssa")
library(Rssa)
plot(co2, ylab=expression("Atmospheric concentration of CO"[2]), las=1)
title(main = "co2 data set")
N=length(co2)
L=round(N/3)
s = ssa(co2,L) # Perform the decomposition
summary(s) # Show various information about the decomposition
dev.new(); plot(s) # Show the plot of the eigenvalues
# Reconstruct into 2 series
r = reconstruct(s, groups=list(Trend=c(1, 4), Seasonality=c(2:3, 5:6)))
dev.new(); plot(r, add.original = TRUE) # Plot the reconstruction
# Calculate 160-point forecast using first 6 components as a base
f = forecast(s, groups=list(1:6), method="bootstrap-recurrent",
len=160, R=20)
dev.new(); # Plot the forecast including the original series
plot(f, shadecols = "green", type = "l")
```

R-code for airpass time series

®

```
airpass=read.csv("http://ssa.cf.ac.uk/ma0367/airpass.csv",
sep=";",header=TRUE); print(airpass)
airpassTS=ts(airpass[,3], start=c(airpass[1,2],1), frequency=12)
plot(airpassTS,type="l", main="airpass data")
points(airpassTS,pch=19,cex=0.6)
N=length(airpassTS)
L=round(N/3)
s = ssa(airpassTS,L) # Perform the decomposition
summary(s) # Show various information about the decomposition
dev.new(); plot(s) # Show the plot of the eigenvalues
# Calculate 70-point forecast using first 5 components as a base
f = forecast(s, groups=list(1:6), method="bootstrap-recurrent", len=70,
R=30)
dev.new(); # Plot the forecast including the N/2 points of original series
plot(f, shadecols = "green", type = "l", include=round(N/2))
lines(f$lower,col="red")
lines(f$upper,col="red") # confidence intervals
```

R-code for Earth Ocean temperature

®

```
data=read.table("http://ssa.cf.ac.uk/ma0367/globtemp.txt",header=TRUE)
print(data[1:4,]); colindex=5;
x=data[,colindex]; xname=names(data)[colindex];
year=data[,1]; month=data[,2];
EOtempTS=ts(x, start=c(year[1],month[1]), frequency=12)
plot(EOtempTS,type="l", main="Earth Ocean temperature")
points(EOtempTS,pch=19,cex=0.6)
N=length(EOtempTS)
L=round(N/8)
s = ssa(EOtempTS,L) # Perform the decomposition
summary(s) # Show various information about the decomposition
dev.new(); plot(s) # Show the plot of the eigenvalues
# Calculate 80-point forecast using first 9 components as a base
f = forecast(s, groups=list(1:9), method="bootstrap-recurrent", len=80,
R=40, base="original")
dev.new(); # Plot the forecast including the N/3 points of original series
plot(f, shadecols = "green", type = "l", include=round(N/3))
lines(f$lower,col="red")
lines(f$upper,col="red") # confidence intervals
```

Rssa package

Several versions of Rssa are available.

The function **ssa(...)** is the same in all versions, but the forecasting function is different: **forecast** or **rforecast/vforecast**.

Please, read the help file via **?forecast**.

```
® f = forecast(s, groups=list(1:9),
method="recurrent", interval = "confidence",
len=80, R=40, base="original")

f = vforecast(s, groups=list(1:10), len=80, only.new
= FALSE, base = "reconstruct")
```

Please, also see 'Fragment 3.5.18' at

`https://ssa-with-r-book.github.io/03-chapter3.html`